Intermittent working-memory coding through NMDA-sustained persistent activity

Joao Barbosa

Cognitive Neuroimaging Unit, INSERM, CEA, CNRS, Université Paris-Saclay, NeuroSpin center & Institut de neuromodulation, GHU Paris, psychiatrie et neurosciences, centre hospitalier Sainte-Anne, pôle hospitalo-universitaire 15, Université Paris Cité, Paris, France.

Abstract

Working memory is classically attributed to persistent spiking sustained by slow NMDA receptor currents. This view has been challenged by large-scale prefrontal recordings in which the remembered item is decoded only intermittently on single trials, suggesting that memories survive silent periods through synaptic facilitation. Here we show that intermittent coding emerges from a canonical attractor network with fixed synapses and a fluctuating, attention-like common input. Brief drops in attention silence the network, but NMDA currents can bridge each silence. Fitted to monkey prefrontal recordings, the model reproduces the number and duration of silent states and, without refitting, predicts how they change with the decoded neurons. We found that silent states are more frequent in less informative samples, validating this prediction and suggesting that measured silences overestimate how often memory leaves spiking activity. Unlike facilitation, the NMDA trace also predicts memory loss when silences outlast it. In line with this, monkeys made more errors when silent states were longer and their performance pointed to a trace of ∼120 ms, matching NMDA kinetics rather than the long time scale of facilitation. Intermittent coding thus reflects persistent activity bridged by NMDA currents.

Introduction

Two main mechanisms have been proposed to hold information in working memory. In persistent-activity models, recurrent excitation keeps a cue-selective population firing through the delay, and slow NMDA receptor currents make this activity stable (Amit and Brunel, 1997; Wang, 1999, 2001; Compte et al., 2000). Consistent with these models, blocking NMDA receptors markedly reduces persistent firing in monkey prefrontal cortex (Wang et al., 2013). In activity-silent models, the memory is instead stored in short-term synaptic facilitation and needs no ongoing spiking during maintenance (Mongillo et al., 2008; Stokes, 2015; Lundqvist et al., 2016; Mi et al., 2017; Miller et al., 2018; Masse et al., 2019; Panichello et al., 2024), and hidden memories can be reactivated after periods without decodable activity (Rose et al., 2016; Wolff et al., 2017). Notably, the two accounts rely on traces with very different time scales. NMDA receptor currents decay on a time scale of hundreds of milliseconds (Vicini et al., 1998; Wang et al., 2008), and models of persistent activity typically use about 100 ms (Wang, 1999; Compte et al., 2000). On the other hand, facilitation at excitatory synapses in prefrontal cortex decays on a time scale of the order of a second (Thomson, 1997; Markram et al., 1998; Hempel et al., 2000; Wang et al., 2006; Mongillo et al., 2008; Mi et al., 2017), and activity-silent models typically use 1.5 s (Mongillo et al., 2008). Synaptic augmentation, which lasts tens of seconds, has been proposed to bridge even longer silences (Mongillo and Tsodyks, 2025).

Large-scale, intracranial recordings now make it possible to test these accounts on single trials. Recent studies have shown that working-memory information in prefrontal cortex fluctuates within a trial (Panichello et al., 2024; Tao and Libedinsky, 2024; Mozumder et al., 2026). Panichello

et al. (2024) found that decoding alternates between *On states* (when the cue can be read out) and *Off states* (when it cannot) and concluded that synaptic changes carry the memory through *Off states*. Tao and Libedinsky (2024) found rare periods of 43–414 ms in which all simultaneously recorded selective neurons were silent for longer than expected by chance, and concluded that short-term plasticity is needed in a very small fraction of trials. Finally, Mozumder et al. (2026) found that persistent activity is asynchronous across neurons, with no population silences beyond chance. In their recordings, decoding *Off states* came with reduced but not abolished tuning and were synchronized between prefrontal and parietal cortex, and the authors interpreted them as fluctuations in the fidelity of a persistent code.

Here, we ask whether a single persistent-activity mechanism can account for these observations and propose that intermittent decoding does not require short-term plasticity and is in fact consistent with the NMDA account. In our model, the memory is held in persistent activity supported by recurrent attractors (Wang, 2002; Wong and Wang, 2006). However, fluctuations in a common input, which we interpret as a fluctuating attentional signal, briefly drop and silence the network. During each silence slow NMDA currents carry the memory across, so the cued population resumes firing when the attentional signal returns. We read out from the model with the same decoding and *On/Off* criteria as Panichello et al. (2024) and fit how often, for how long and how deeply attention lapses to the *On/Off* statistics of these recordings. The model reproduces quantitatively the number and length of *On and Off states* while keeping the memory. Our reanalyses of the recordings showed that *Off states* are more frequent when fewer (or less selective) neurons are decoded, so their frequency partly reflects how well the recorded sample reads out the memory. Without refitting, the attractor model predicts this dependence for decoded populations it was never fitted to. Intermittent decoding is therefore compatible with persistent activity sustained by NMDA currents, without short-term plasticity. Finally, unlike a model in which short-term facilitation carries the memory, the NMDA model predicts that errors grow with the time spent *Off*, as we found in the recordings. Fitted to the monkeys' performance, the decline in accuracy with *Off* time points to a trace of about 120 ms, which matches NMDA kinetics rather than facilitation. Previous work took intermittent decoding as evidence that persistent activity is not sufficient for working memory, but our results show that this evidence has a simpler explanation within the standard persistent-activity account.

Results

Input dips produce *Off states* and NMDA carries the memory across them

The model is a classic two-pool attractor network (Wang, 2002; Wong and Wang, 2006), with two memory pools, cued and uncued, that excite themselves through NMDA currents and compete through shared inhibition (Methods). Without fluctuations in its input, the cued pool stays active after the cue and decoding stays *On* (Fig. 1a, first second after the cue). The key addition to this classic model is a common input to both pools that fluctuates, which we interpret as an attentional signal, since attention is known to wax and wane on sub-second time scales (Fiebelkorn and Kastner, 2019; see Discussion). With brief dips in this input, both pools fall silent and decoding drops to chance (Fig. 1a, after 1,000 ms). During each dip, the NMDA gating of the cued pool decays with a time constant of 150 ms and stays higher than that of the uncued pool. When the input returns, the pool with more NMDA gating wins the competition. *Off states* therefore do not erase the memory, because NMDA gating carries it across each silence, and the cue is still correctly remembered at the end of the delay.

To compare the model with the data, we read it out like the recordings (Supplementary Fig. 1; Methods). In the data, a decoder trained on the recorded neurons gives, every 10 ms, a z score of how well the cue can be read out. The model has no neurons, so we built the equivalent score from its two pools. The evidence for the cue is the difference between the cued and uncued pool rates in the same 100 ms windows. A gain converts this difference into z units, and read-out

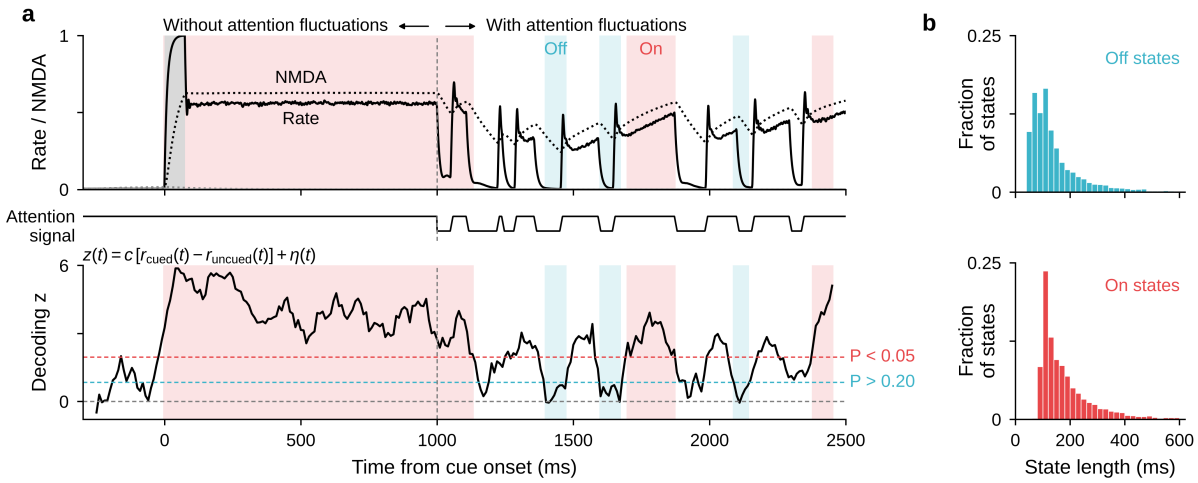


Figure 1: Input dips create *On and Off states*, and NMDA carries the memory. a, Example trial. Top: rates of the cued (black) and uncued (grey) pools, relative to the maximum rate, and their NMDA gating (dotted); grey shading marks the cue. Middle: common input (attention signal), constant for the first 1,000 ms after the cue and with dips afterwards. The dip-free first second is shown for illustration only; in all other simulated trials, dips occur throughout the delay. Bottom: decoding z score, with *On* (red) and *Off* (cyan) states. Dashed lines: *On* threshold (z = 1.96) and *Off* threshold (z = 0.84), as in Panichello et al. (2024). **b**, Length distributions of *Off* (top) and *On* (bottom) states in the model.

noise stands for the variability of a decoder trained on a limited sample of neurons, with the same variance and temporal correlations as the decoder output before the cue. Importantly, the gain was not fitted to the *On/Off* statistics. Instead, it was calibrated on a single quantity measured in the data, the typical z value when decoding is high, which we call the decoding strength (Methods). When synthetic neurons driven by the model were decoded with the same pipeline as the data, the decoder's z agreed with this read-out as closely as two runs of the decoder on independent spikes, and gave similar *Off states* (Supplementary Fig. 1f–j). *On and Off states* (red and cyan in Fig. 1a) are then labelled with the thresholds of Panichello et al. (2024) (dashed lines in Fig. 1a). Read out this way, the model alternates between *On and Off states* (Fig. 1a, bottom). Both length distributions peak near 100 ms and have long tails (Fig. 1b), qualitatively like those in the recordings (Panichello et al., 2024). In the next sections, we ask whether the model also matches the data quantitatively.

On/Off statistics depend on which neurons are decoded

So far we considered decoders that use all recorded neurons. We found, however, that *On/Off* statistics depend strongly on which neurons the decoder uses, which we call the decoded population (Fig. 2, black circles). Decoders using the most cue-selective neurons spend much more of the delay *On*, and less *Off*, than decoders using the same numbers of randomly chosen neurons, which also show more *Off states* per trial (paired Wilcoxon signed-rank tests over 25 sessions, $p < 10^{-6}$ for each of 5, 20 and 50 neurons; Fig. 2). The decoded populations differ mainly in decoding strength, which is much higher for the most selective neurons. How often a memory appears *Off* therefore depends partly on how well the recorded neurons read it out. This matters for interpreting *Off states* as evidence for activity-silent memory: a decoder trained on a limited sample of neurons can fall to chance while the memory is still present in the activity of other, potentially unrecorded neurons. The measured fraction of time *Off* is therefore an upper bound on how often the memory leaves spiking activity. We then asked if our model could explain this dependence.

To test this, we fitted the dynamics only to the decoder using all neurons in 25 sessions

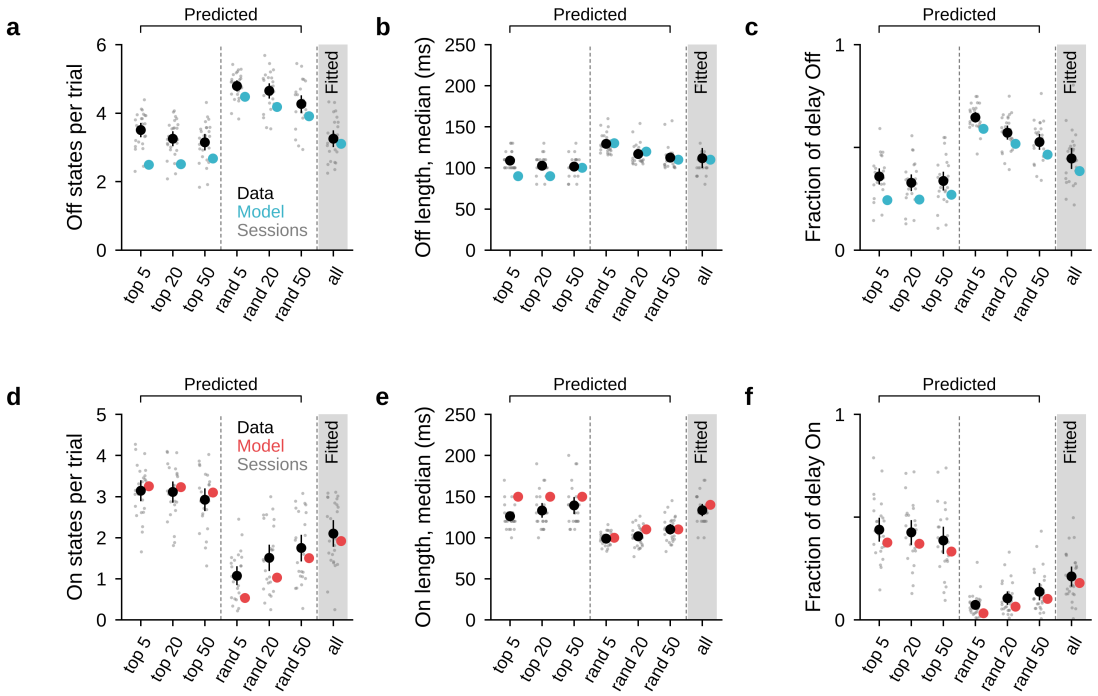


Figure 2: The model predicts how *On/Off* statistics change with the decoded population. Decoders using the most cue-selective neurons (top 5, 20, 50) are *On* more often and *Off* less often than decoders using the same numbers of random neurons (rand 5, 20, 50). Black circles: data, mean ± 95% CI (2 SEM) over 25 sessions. Grey dots: single sessions. Cyan and red circles: model, *Off and On* statistics. The model was fitted to all neurons only (grey band); the other six decoded populations are predictions obtained by changing only the read-out gain and noise.

(Methods). The model matches the data well in *Off states* per trial (3.1 vs 3.3), *Off* median length (110 vs 112 ms), *On states* per trial (1.9 vs 2.1) and *On* median length (140 vs 134 ms) (Fig. 2, grey band). We then kept the dynamics fixed and, for each of the six other decoded populations, changed only the read-out without refitting any parameters. Its gain was calibrated on that population's decoding strength and its noise on that population's pre-cue z (Methods). The model thus predicts quantitatively the shift between selective and random neurons (Fig. 2; Supplementary Fig. 2).

Errors increase with time *Off*, as expected from an NMDA trace

So far, we analysed correct trials only, but error trials can help distinguish between the two accounts of working memory. NMDA currents and short-term synaptic facilitation can both carry a memory across brief *Off states*, but on very different time scales. In our model, NMDA gating decays with a time constant of 150 ms, whereas facilitation decays on a time scale of the order of a second (Wang et al., 2006; Mongillo et al., 2008), comparable to the length of the whole memory period (the delay ended 1.4–1.6 s after cue onset). To see what each mechanism predicts, we compared our model with a version in which facilitation replaces NMDA gating (STP model; Methods). Fitted in the same way, it reproduced the *On/Off* statistics about as well as the NMDA model (Supplementary Fig. 3). We then silenced each model with a single dip of the common input of increasing duration (single-dip test; Methods). The NMDA model held the memory for silences of up to 350 ms and lost it for every silence from 400 ms on, because the NMDA trace had decayed too far for the cued pool to restart (about 2.5 τ_N), whereas the STP model held the memory for silences as long as 1 s (Fig. 3a). The NMDA account therefore predicts that errors become more frequent when the memory stays *Off* for long, whereas facilitation

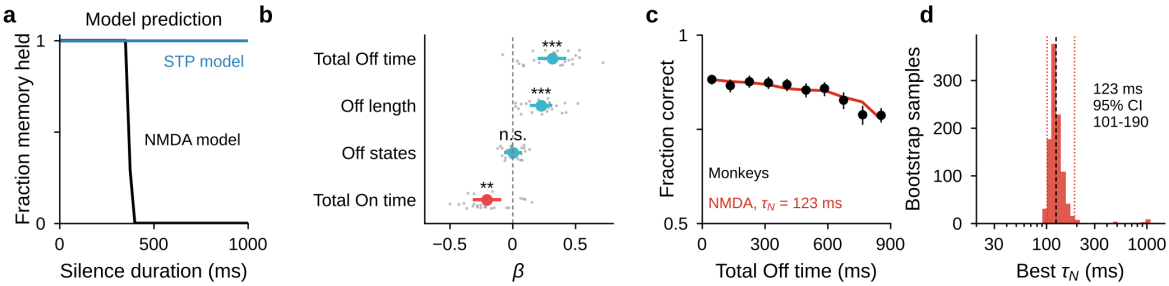


Figure 3: Errors increase with time *Off*, and the decline in performance points to a trace of about 120 ms. a, Model prediction (single-dip test): fraction of trials with the memory held against the duration of a single silence, imposed by one dip of the common input (0 to 1 s in 25 ms steps, 300 trials per duration), for the NMDA model (black) and the STP model (blue). Memory was scored on the network, without read-out (Methods). Shading: 95% Wilson interval. **b**, Effect on the log-odds of an error (β) per within-session SD of four statistics of the delay (500–1,400 ms after cue onset): total *Off* time, mean *Off-state* length, number of *Off states* and total *On* time. Coloured circles and lines: fixed effect and 95% CI of a logistic mixed model, fitted for each statistic to all 25 sessions (Methods). Grey dots: single sessions (fixed effect plus the session's random slope). Likelihood-ratio tests: total *Off* time, $p = 1{\times}10^{-5}$; *Off* length, $p = 7{\times}10^{-5}$; number of *Off states*, $p = 0.89$; total *On* time, $p = 0.001$. $^{***}p < 0.001$, $^{**}p < 0.01$, n.s. not significant. **c**, Fraction correct against total *Off* time in the delay, all 15,468 trials pooled, in 90 ms bins (black; 95% Wilson interval), and the NMDA model with the best-fitting τ_N of 123 ms (red), linked to behaviour by P(correct) = p0 × [P(held) + P(lost)/8] (Methods). **d**, Best τ_N in 1,000 bootstraps over sessions. Dashed: fit to all data. Dotted: 95% interval (101–190 ms).

predicts no such relation for silences shorter than its own decay, which here spans the whole delay.

We tested this prediction by comparing *On/Off* statistics between correct and error trials (Methods). Within sessions, the odds of an error increased with the total *Off* time in the delay and with the mean length of *Off states*, and decreased with the total *On* time, but did not depend on the number of *Off states* (Fig. 3b). Errors thus depended on how long the memory was *Off*, not on how often it went *Off*: frequent *Off states* are harmless as long as each is brief. Across all 25 sessions, total *Off* time increased the odds of an error ($\beta = 0.32$, 95% CI 0.20–0.43, $p = 1 \times 10^{-5}$; logistic mixed model with sessions as random effects, Methods). The effect was clear in monkeys H and J ($\beta = 0.48$, $p = 1 \times 10^{-4}$, and $\beta = 0.52$, $p = 0.005$) but weak in monkey A ($\beta = 0.10$, $p = 0.13$).

Finally, we asked which time constant of the trace best explains how performance falls with *Off* time. We varied τ_N in the fitted NMDA model, keeping the dip statistics and all other parameters unchanged, and simulated each version with the same random dips, read out as the data. In these simulations, the memory is lost when a single *Off* period outlasts the NMDA trace and the cued pool cannot restart (with $\tau_N = 150$ ms, single dips longer than about 375 ms; Fig. 3a), whereas long *Off* times made of several brief *Off states* are mostly bridged. We linked the probability that the memory was held to the probability of a correct response, assuming that the monkeys guess among the 8 locations after a memory loss and that other errors do not depend on *Off* time (Methods). With the best-fitting τ_N of 123 ms, the model reproduced the decline of the pooled fraction correct (Fig. 3c). Resampling sessions gave a 95% confidence interval of 101–190 ms (Fig. 3d). The estimate also hardly depended on what the monkeys do after losing the memory. Assuming that they are always wrong, or guess correctly up to half of the time, gave 102–129 ms (Methods). The decline of performance with *Off* time thus calls for a trace of about 120 ms, close to measured NMDA receptor kinetics and to the 150 ms assumed in the fitted model, and far shorter than the time scale of facilitation, of the order of a second (Thomson, 1997; Markram et al., 1998; Hempel et al., 2000; Wang et al., 2006; Mongillo et al., 2008).

Discussion

Recent large-scale recordings found that prefrontal decoding of a remembered item is intermittent on single trials (Panichello et al., 2024; Tao and Libedinsky, 2024; Mozumder et al., 2026), suggesting that memories survive silent periods through synaptic facilitation (Mongillo et al., 2008; Stokes, 2015; Lundqvist et al., 2016; Mi et al., 2017; Miller et al., 2018; Masse et al., 2019; Panichello et al., 2024). We showed that the same *On/Off* statistics arise in a conventional attractor network with fixed synapses. In the model, *Off states* occur when a common input briefly drops, and the memory survives each silence because NMDA receptor currents decay slowly enough to bias which pool restarts. Fitted to the decoder trained on the empirical data, the model reproduced the number and length of *On and Off states* while keeping the memory. Unlike a model in which short-term facilitation carries the memory, it also predicted that errors grow with the time spent *Off*, which we validated in prefrontal recordings.

In the recordings, *Off states* were more frequent and *On states* rarer when the decoder used random rather than selective neurons. This showed that a decoder falls to chance whenever the memory signal in its sample is weak relative to its noise, not only when the memory is absent from the network. The fraction of time *Off* measured from a limited sample is therefore an upper bound on how often the memory leaves spiking activity, and comparisons of *Off-state* statistics across studies or brain areas need to take the decoded population into account. The model captured this shift without refitting by changing only the read-out gain and noise, both calibrated from the data.

Our account is consistent with the other recent recordings of *Off states*. Mozumder et al. (2026) found that *Off states* were synchronized between prefrontal and posterior parietal cortex, came with lower firing rates, and showed no obvious rhythmicity. A common input that reaches several areas and fluctuates irregularly, as in our fit, would produce such coordinated, aperiodic *Off states*. Tao and Libedinsky (2024) likewise found that information is reduced but present between gamma bursts. Our model is compatible with both, because NMDA currents can carry the memory across full silences but are not needed when activity is only reduced. While attention is sustained, the memory is held by persistent activity without any *Off state* (Fig. 1a, first second after the cue).

We interpreted the common input as an attentional signal that is high when attention is deployed on the memorandum and drops during lapses. Two lines of evidence support this interpretation. First, Engel et al. (2016) found that spiking in macaque area V4 alternates between *On and Off* episodes that occur synchronously across cortical layers. These states were identified directly from population spiking, not from a decoder. *Off* episodes lasted about 100 ms on average (102 ms during stimulus-driven and 118 ms during spontaneous activity), close to the *Off states* in prefrontal decoding (about 110 ms) and within the time an NMDA trace can bridge. Spatial attention lengthened *On* episodes and made transitions less frequent, and monkeys detected a stimulus change more often when it occurred during an *On* episode than during an *Off* episode (78% vs 65%). Second, sustained attention samples the environment rhythmically rather than continuously. In humans, detection of threshold stimuli depends on the phase of ongoing EEG oscillations near 7 Hz (Busch et al., 2009), especially at attended locations (Busch and VanRullen, 2010). When attention is divided between two locations, detection at each fluctuates at about 4 Hz (Landau and Fries, 2012). In monkeys, frontal eye field and lateral intraparietal activity alternates at theta frequencies (3–8 Hz) between periods of enhanced perception and periods in which attention is more likely to shift (Fiebelkorn et al., 2018), and human intracranial recordings show similar theta-rhythmic sampling in frontoparietal cortex (Helfrich et al., 2018). Because attention and working memory share prefrontal mechanisms (Awh and Jonides, 2001; Panichello and Buschman, 2021), lapses of attention could plausibly reach the memory network. In the fitted model, the attention signal recovers about five times per second, or one cycle every ∼200 ms. This is in the low theta range reported for attentional sampling (Landau and Fries, 2012; Fiebelkorn et al., 2018; Helfrich et al., 2018) and only slightly slower

than the ∼7 Hz phase effects on detection (Busch et al., 2009; Busch and VanRullen, 2010). However, the fitted signal was not periodic. A direct test would be to ask whether *Off-state* onsets lock to the phase of theta activity in the prefrontal local field potential, or whether probes presented during *Off states* are detected less often.

In our model, the memory is carried across each silence by NMDA gating that decays with a time constant of 150 ms and reflects recent firing rather than plasticity. This lies within the range of NMDA receptor currents, which decay over hundreds of milliseconds (Vicini et al., 1998; Wang et al., 2008) and are required for persistent firing in primate prefrontal cortex (Wang et al., 2013). *Off states* last about 100 ms, within the time an NMDA trace can bridge, so no longer-lasting mechanism is needed to explain them. The behavioural data point the same way. A memory held by facilitation decaying on a time scale of the order of a second should hardly be affected by a few hundred more milliseconds *Off*, as the STP model showed. We found that errors increased with the time spent *Off* and with the length of *Off states*, but not with their number (Fig. 3). This is expected from a trace that bridges brief silences but not long ones. Fitted to the monkeys' performance, the time constant of this trace was about 120 ms (95% CI 101–190 ms), close to measured NMDA receptor kinetics and far shorter than that of facilitation. Activity-silent mechanisms may still matter on longer time scales, for example to carry information between trials, when prefrontal memory traces are reactivated before the next stimulus (Barbosa et al., 2020). Notably, this between-trial trace is not a voluntary memory. It does not require attention and shows up as involuntary serial biases in behaviour, rather than as the information the subject actively maintains (Kiyonaga et al., 2017; Pascucci et al., 2023). Persistent activity and slower silent traces are therefore not mutually exclusive.

In sum, intermittent decoding of working memory is expected from an attractor network with slow NMDA currents and a fluctuating, attention-like input. Its statistics depend on how well the recorded neurons read out the memory, and errors increase with the time spent *Off*, as expected from the limited duration of the NMDA trace. These results reconcile intermittent rate coding with persistent-activity accounts identify a trace of about 120 ms that bridges *Off states*, and link *Off states* to the rhythmic but irregular nature of attentional sampling.

Methods

Rate model

The network is the reduced two-pool attractor model of Wong and Wang (2006), with the changes described below. Two excitatory memory pools, $i \in \{\text{cued}, \text{uncued}\}$, have rates r_i and NMDA gating s_i. They share an inhibitory population r_I and receive a feed-forward inhibition q that follows the common input $m(t)$ with a lag. Parameters in bold were free during fitting; all others were fixed.

$$\tau_r \frac{dr_i}{dt} = -r_i + \varphi(x_i), \tag{1}$$

$$x_i = \boldsymbol{w_s} s_i + w_a r_i - w_I r_I + I_0\, m(t) - w_{ff}\, q + \text{cue}_i + \boldsymbol{\sigma}\xi_i, \tag{2}$$

$$\tau_N \frac{ds_i}{dt} = -s_i + \beta\, r_i(1 - s_i), \tag{3}$$

$$\tau_I \frac{dr_I}{dt} = -r_I + w_{EI}\,(r_{\text{cued}} + r_{\text{uncued}})\,, \tag{4}$$

$$\tau_{ff} \frac{dq}{dt} = -q + m(t). \tag{5}$$

The transfer function φ is a logistic sigmoid, $\varphi(x) = r_{\max}/\big(1 + \exp(-(x-1)/\kappa)\big)$, with maximum rate $r_{\max}$, threshold 1 and slope κ, and ξ_i are independent Ornstein–Uhlenbeck noises ($\tau_\xi = 10$ ms). Fixed parameters: $r_{\max} = 150$, $\kappa = 0.06$, $w_a = 0.0141$, $\beta = 0.02$, $w_I = 0.04$, $w_{EI} = 0.5$, $w_{ff} = 0.8$, $I_0 = 1.496$; $\tau_r = 10$ ms, $\tau_I = 1$ ms, $\tau_{ff} = 15$ ms, $\tau_N = 150$ ms. The implementation differs

from Wong and Wang (2006) in three respects: the transfer function is a saturating sigmoid, competition between the pools goes through an explicit inhibitory population, and a lagging feed-forward inhibition is added. Parameter values could therefore not be taken directly from their study. We chose them by hand so that each pool is bistable, with a low state near 0 Hz and a high state of about 85 Hz, and so that shared inhibition allows only one pool to be active at a time. The cue is a 50 ms input of amplitude 1 to the cued pool, after 300 ms of baseline, followed by a 1,600 ms delay. Equations were integrated with Euler steps of 0.1 ms. The memory was scored correct when, in the last 100 ms, the NMDA gating of the cued pool exceeded the uncued pool's by more than 0.05. We scored NMDA gating rather than rates because the network may be silent at the end of the delay, and a strong attentional input, modelling the response period, restarts the pool with more NMDA gating and so recovers the correct response (not shown).

Our main modification to this model is the common input $m(t)$ to both pools, which represents fluctuating attention. It is 1 except during dips, where it drops to $1 - \boldsymbol{\delta}$ with 10 ms ramps. Dip lengths are gamma-distributed (mean $\boldsymbol{\mu_d}$, shape $\boldsymbol{k}$), and the gaps between dips are exponential with mean $1/\boldsymbol{\lambda_d}$.

Data

We reanalysed the recordings of Panichello et al. (2024): 25 sessions, correct trials only (error trials were used only in the behavioural analysis), all cue locations (13,190 trials; 329 units per session on average). For each time bin, an L2-regularised logistic decoder (λ = 10,000) was evaluated with approximate leave-one-out and z-scored against 30 shuffled label sets per cue pair. Decoded populations were all neurons, the k most cue-selective neurons (ranked by split-half ANOVA), and k randomly chosen neurons (10 draws, averaged within sessions), with k = 5, 20 and 50. Decoded populations of the same size were compared within sessions with two-sided Wilcoxon signed-rank tests on the session values (random draws averaged within each session).

Behavioural analysis

To relate *On/Off states* to behaviour, we also labelled error trials (15,468 trials in total, 2,278 errors). Error trials were decoded with decoders fitted on all correct trials of the same cue pair, so they were also out of sample, and their rates were z-scored with the mean and SD of correct trials. For each trial, we computed the total *Off* time and the total *On* time in the delay (500–1,400 ms after cue onset), the number of *Off states* and their mean length (13,962 trials with at least one *Off state*). For each statistic, the probability of an error was modelled with a logistic mixed model, error $\sim$ z(stat) + t + t^2 + monkey + (1 + z(stat) || session), where z(stat) is the statistic z-scored within each session, so that only within-session variation is used, t is the time within the session, and each session has a random intercept and an uncorrelated random slope. Monkey was a fixed effect, as three monkeys are too few to estimate a random effect. Models were fitted with the Laplace approximation and tested with likelihood-ratio tests against the same model without the statistic. For the fit of τ_N, trials were pooled over sessions and binned by *Off* time in 90 ms bins (one tenth of the delay window).

STP model

To compare our model with a short-term plasticity account, we replaced the NMDA gating s_i in the recurrent input by a saturating synaptic activation without a time constant of its own, $S_i = g_i/(1 + g_i)$, with $g_i = \beta r_i u_i / U$, where u_i is Tsodyks–Markram short-term facilitation (Mongillo et al., 2008; Mi et al., 2017):

$$\tau_f \frac{du_i}{dt} = U - u_i + \tau_f U(1 - u_i) r_i,$$

with $U = 0.3$ and $\tau_f = 1.5$ s. To simplify fitting and to give the STP model the best chance to fit the data, synaptic depression was switched off: with depression ($\tau_d = 0.3$ s), each re-ignition ended too early and *On states* were too short (112 vs 172 ms in the data). All other parameters were as in the NMDA model, and the STP model was fitted in the same way to the decoder using all neurons, with an additional penalty for population spikes before the cue. Best fit: $\boldsymbol{w_s} = 0.69$, $\boldsymbol{\lambda_d} = 5.5$/s, $\boldsymbol{\mu_d} = 212$ ms, $\boldsymbol{k} = 0.70$, $\boldsymbol{\delta} = 0.39$, $\boldsymbol{\sigma} = 0.005$. Both models matched the *On/Off* statistics about equally well (loss 9.7 for the NMDA model and 10.4 for the STP model, on 2,112 trials) and held the memory in nearly all trials.

Simulations of memory against *Off* time

To compare with the behaviour, each fitted model was then simulated with random dips drawn from its fitted dip statistics (100,000 trials per model); the memory was held in 98.5% (NMDA model) and 99.99% (STP model) of these trials. To measure how long each model can hold the memory through a single silence (single-dip test, Fig. 3a), we simulated trials with only one dip of the common input, starting 500 ms after cue onset, with the model's fitted depth and a length from 0 to 1,000 ms in 25 ms steps (300 trials per length, 12,300 trials per model). Fig. 3a shows the fraction of these trials with the memory held against dip duration. Trials were read out with the all-units gain and noise of the fitted model, and the memory was scored as in the NMDA model, using the facilitation u_i instead of the NMDA gating in the STP model. *Off* time was computed as in the data.

NMDA time constant fitted to performance

The fitted NMDA model, with random dips drawn from the fitted dip statistics (the same dips for every τ_N) and the all-units read-out, was simulated with τ_N from 20 to 1,500 ms (17 values, 30,000 trials each) and all other parameters unchanged, so the *On/Off* statistics drift away from their fitted values as τ_N changes. For each τ_N, the probability that the memory was held was computed in the ten *Off*-time bins and interpolated in log τ_N. The model was linked to behaviour by P(correct) = $p_0 \times$ [P(held) + G $\times$ P(lost)], where G = 1/8 is the chance of a correct guess among the 8 cue locations after the memory is lost, and p_0 is the performance when the memory is held, which absorbs errors that do not depend on *Off* time. τ_N and p_0 were fitted by maximum likelihood to all 15,468 trials pooled (best $p_0 = 0.887$). Confidence intervals came from 1,000 bootstraps over sessions (25 sessions drawn with replacement, their trials pooled and refitted). The estimate depended little on G: the best τ_N was 129, 118 and 102 ms for G = 0, 1/4 and 1/2.

Fitting

Six parameters were fitted with the covariance matrix adaptation evolution strategy (CMA-ES; Hansen and Ostermeier, 2001) to the decoder using all neurons: $\boldsymbol{w_s}$, $\boldsymbol{\lambda_d}$, $\boldsymbol{\mu_d}$, $\boldsymbol{k}$, $\boldsymbol{\delta}$ and $\boldsymbol{\sigma}$. The NMDA decay time was not fitted but fixed at $\tau_N = 150$ ms, within the range measured in cortex (Vicini et al., 1998; Wang et al., 2013). The loss combined the squared scaled errors of six statistics (*On and Off states* per trial, mean lengths and fractions of the delay), the squared Hellinger distance between model and data length distributions (states ≥ 40 ms), and a penalty when the memory was correct in fewer than 98% of trials. Best fit: $\boldsymbol{w_s} = 1.30$, $\boldsymbol{\lambda_d} = 10.2$/s, $\boldsymbol{\mu_d} = 90$ ms, $\boldsymbol{k} = 8.7$, $\boldsymbol{\delta} = 0.60$, $\boldsymbol{\sigma} = 0.006$.

Decoding read-out, *On/Off* labelling and statistics

To compare the model with the recordings, we measured *On* and *Off states* as close as possible to how they were measured in the data (Panichello et al., 2024), where they are defined on the z score of a decoder trained on the neurons recorded in a session. The model does not generate the

activity of individual neurons. Instead, it describes the latent dynamics of a working-memory circuit, from which some neurons were recorded. Labelling the pool rates of the model directly would ignore the read-out noise inherent to this subsampling, and the dependence of *On/Off* statistics on how well the recorded neurons read out the memory (Fig. 2). We therefore converted the model's activity into a decoding z score with the same time resolution, noise and labelling as the data (Supplementary Fig. 1).

At each time step, the decoding evidence was $\Delta r(t)$, the difference between the cued and uncued pool rates. Δr is large when the cued pool is active and near zero when both pools are silent, and it was averaged in 100 ms windows stepped by 10 ms, as in Panichello et al. (2024). We modelled the z score as $z(t) = c\,\Delta r(t) + \eta(t)$, where the gain c converts rates into z units and η is the read-out noise, which stands for the variability of a decoder trained on a finite number of neurons and trials (Supplementary Fig. 1a).

The noise η was estimated from the data before the cue, where the decoder z carries no memory signal and therefore reflects read-out noise alone. Using the pre-cue data, we matched two of its properties, the noise variance and temporal correlation. The standard deviation of pre-cue z was 1.06, close to the value of 1 expected for a z score, so we gave η unit variance. In Panichello et al. (2024), z was computed in 100 ms windows stepped by 10 ms, so consecutive points share most of their spikes and the noise at two points Δt apart is correlated by $1 - \Delta t/100$ ms for $\Delta t < 100$ ms. We generated this part of η as white noise averaged in the same overlapping windows, $\varepsilon(t) = \frac{1}{\sqrt{10}} \sum_{j=0}^{9} w(t - 10j \text{ ms})$, where w are independent standard normal values drawn every 10 ms. In the data, z remained correlated beyond 100 ms (Supplementary Fig. 1c), possibly because the decoder output also drifts slowly from trial to trial. We captured this drift with a constant offset added to each trial (ξ), whose share of the variance (ρ) was set to the correlation of pre-cue z between time points 200 ms apart, which is far enough that the windows do not overlap ($\rho = 0.35$ for all neurons, 0.09 for the most selective neurons and 0.11 for random neurons). Thus $\eta(t) = \sqrt{1-\rho}\,\varepsilon(t) + \sqrt{\rho}\,\xi$, where ε is the window-averaged white noise and ξ the per-trial offset, both with unit variance. The autocorrelation of η is then $(1-\rho)(1 - \Delta t/100 \text{ ms}) + \rho$ for $\Delta t < 100$ ms and ρ beyond, as in the data (Supplementary Fig. 1c).

We also calibrated the gain c on a separate property of the data, the level that z reaches when the memory is read out, which we call the decoding strength. The mean of the delay z would not serve this purpose, because the delay z mixes time points with strong and weak decoding (e.g. *On* vs *Off*). We therefore fitted a two-component Gaussian mixture to the delay z values (500–1,400 ms after cue onset) and took the mean of the upper component, which reflects the level of z during strong decoding (*On*) and depends much less on how often z is low (*Off*). For each decoded population, c was set so that the decoding strength of the model's delay z equaled the data's, averaged over sessions (Supplementary Fig. 1d). Decoding strength increases steeply and monotonically with c, so c is well determined (Supplementary Fig. 1e).

States were labelled exactly as in Panichello et al. (2024), so that model and data are compared with identical definitions. *On states* are runs of $z > 1.96$ whose summed z exceeds the 95th percentile of the same quantity in pure noise η. *Off states* are runs of at least 5 points (50 ms) with $z < 0.84$. States were counted from 0 to 1,400 ms after cue onset, and fractions of the delay spent *On* or *Off* from 500 to 1,400 ms. Model statistics were computed from 2,112 simulated trials, four times the average number of correct trials per session (528), so that they are estimated more precisely than in any single session. During fitting, each evaluation used 792 trials (one and a half sessions) to keep the optimization fast. Data statistics are means over the 25 sessions; for random neurons, the 10 draws were first averaged within each session.

Our read-out approximates a real decoder by a simple formula ($z(t) = c\,\Delta r(t) + \eta(t)$), so we tested whether the calibration recovers the gain and the *On/Off* statistics of a real decoder in synthetic data where the ground truth is known (Supplementary Fig. 1f–j). We used the fitted model to generate spikes and decoded them with the unchanged data pipeline. The model was simulated for two cue locations, A and B. When A was cued, pool A was the cued pool and

pool B the uncued one, and the reverse when B was cued, which gives the decoder two classes to discriminate, as in the data. The pool rates drove 330 Poisson neurons, the average number of units recorded per session (329). 30% of them followed pool A or pool B (half each) and the rest were untuned. Baseline rates were lognormal with a median of 5 Hz, and all neurons shared a per-trial gain, which produces slow trial-to-trial drift as in the data. We simulated 70 trials per cue location, close to the number of correct trials per location in a session, and 8 synthetic sessions at each of three tuning strengths, chosen so that the decoder's decoding strength matched that of the data's random-20, all-unit and top-20 decoders (Supplementary Fig. 1g). The fraction of tuned neurons matters only through this decoding strength, which the tuning strength sets. We then applied to the synthetic decoder output exactly the calibration used for the data. The recovered gain c grew in proportion to the tuning strength of the synthetic neurons, and its values (0.030, 0.050 and 0.086) were close to the gains of the model calibrated on the corresponding data decoders (0.029, 0.051 and 0.088; Supplementary Fig. 1h). With this gain and with ρ estimated from the pre-cue autocorrelation, the read-out z correlated with the decoder's as closely as two runs of the decoder on independent spikes from the same rates (Supplementary Fig. 1i). It reproduced the number and fraction of *Off states*, and undercounted *On states* by about one per trial (Supplementary Fig. 1j).

References

Daniel J. Amit and Nicolas Brunel. Model of global spontaneous activity and local structured activity during delay periods in the cerebral cortex. *Cerebral Cortex*, 7:237–252, 1997.

Edward Awh and John Jonides. Overlapping mechanisms of attention and spatial working memory. *Trends in Cognitive Sciences*, 5:119–126, 2001.

Joao Barbosa et al. Interplay between persistent activity and activity-silent dynamics in the prefrontal cortex underlies serial biases in working memory. *Nature Neuroscience*, 23:1016–1024, 2020.

Niko A. Busch and Rufin VanRullen. Spontaneous EEG oscillations reveal periodic sampling of visual attention. *Proceedings of the National Academy of Sciences USA*, 107:16048–16053, 2010.

Niko A. Busch, Julien Dubois, and Rufin VanRullen. The phase of ongoing EEG oscillations predicts visual perception. *Journal of Neuroscience*, 29:7869–7876, 2009.

Albert Compte, Nicolas Brunel, Patricia S. Goldman-Rakic, and Xiao-Jing Wang. Synaptic mechanisms and network dynamics underlying spatial working memory in a cortical network model. *Cerebral Cortex*, 10:910–923, 2000.

Tatiana A. Engel, Nicholas A. Steinmetz, Marc A. Gieselmann, Alexander Thiele, Tirin Moore, and Kwabena Boahen. Selective modulation of cortical state during spatial attention. *Science*, 354:1140–1144, 2016.

Ian C. Fiebelkorn and Sabine Kastner. A rhythmic theory of attention. *Trends in Cognitive Sciences*, 23:87–101, 2019.

Ian C. Fiebelkorn, Mark A. Pinsk, and Sabine Kastner. A dynamic interplay within the frontoparietal network underlies rhythmic spatial attention. *Neuron*, 99:842–853, 2018.

Nikolaus Hansen and Andreas Ostermeier. Completely derandomized self-adaptation in evolution strategies. *Evolutionary Computation*, 9:159–195, 2001.

Randolph F. Helfrich et al. Neural mechanisms of sustained attention are rhythmic. *Neuron*, 99: 854–865, 2018.

Chris M. Hempel, Kenichi H. Hartman, Xiao-Jing Wang, Gina G. Turrigiano, and Sacha B. Nelson. Multiple forms of short-term plasticity at excitatory synapses in rat medial prefrontal cortex. *J Neurophysiol*, 83:3031–3041, 2000.

Anastasia Kiyonaga, Jason M. Scimeca, Daniel P. Bliss, and David Whitney. Serial dependence across perception, attention, and memory. *Trends Cogn Sci*, 21:493–497, 2017.

Ayelet N. Landau and Pascal Fries. Attention samples stimuli rhythmically. *Current Biology*, 22: 1000–1004, 2012.

Mikael Lundqvist, Jonas Rose, Pawel Herman, Scott L. Brincat, Timothy J. Buschman, and Earl K. Miller. Gamma and beta bursts underlie working memory. *Neuron*, 90:152–164, 2016.

Henry Markram, Yun Wang, and Misha Tsodyks. Differential signaling via the same axon of neocortical pyramidal neurons. *Proc Natl Acad Sci USA*, 95:5323–5328, 1998.

Nicolas Y. Masse, Guangyu R. Yang, H. Francis Song, Xiao-Jing Wang, and David J. Freedman. Circuit mechanisms for the maintenance and manipulation of information in working memory. *Nat Neurosci*, 22:1159–1167, 2019.

Yuanyuan Mi, Mikhail Katkov, and Misha Tsodyks. Synaptic correlates of working memory capacity. *Neuron*, 93:323–330, 2017.

Earl K. Miller, Mikael Lundqvist, and André M. Bastos. Working memory 2.0. *Neuron*, 100: 463–475, 2018.

Gianluigi Mongillo and Misha Tsodyks. Synaptic encoding of time in working memory. *eLife*, 14: RP107005, 2025.

Gianluigi Mongillo, Omri Barak, and Misha Tsodyks. Synaptic theory of working memory. *Science*, 319:1543–1546, 2008.

Rana Mozumder, Zhengyang Wang, Wenhao Dang, Junda Zhu, Benjamin M. Hammond, Anna Machado, and Christos Constantinidis. Asynchronous firing and off states in working memory maintenance. *Cell Reports*, 45:116764, 2026.

Matthew F. Panichello and Timothy J. Buschman. Shared mechanisms underlie the control of working memory and attention. *Nature*, 592:601–605, 2021.

Matthew F. Panichello et al. Intermittent rate coding and cue-specific ensembles support working memory. *Nature*, 636:422–429, 2024.

David Pascucci, Ömer Dağlar Tanrikulu, Ayberk Ozkirli, Christian Houborg, Gizay Ceylan, Paul Zerr, Mohsen Rafiei, and Árni Kristjánsson. Serial dependence in visual perception: a review. *J Vis*, 23(1):9, 2023.

Nathan S. Rose, Joshua J. LaRocque, Adam C. Riggall, Olivia Gosseries, Michael J. Starrett, Emma E. Meyering, and Bradley R. Postle. Reactivation of latent working memories with transcranial magnetic stimulation. *Science*, 354:1136–1139, 2016.

Mark G. Stokes. 'activity-silent' working memory in prefrontal cortex: a dynamic coding framework. *Trends in Cognitive Sciences*, 19:394–405, 2015.

Weicheng Tao and Camilo Libedinsky. Evidence of activity-silent working memory in prefrontal cortex. *bioRxiv*, 2024. doi:10.1101/2024.06.03.597259.

Alex M. Thomson. Activity-dependent properties of synaptic transmission at two classes of connections made by rat neocortical pyramidal axons in vitro. *J Physiol*, 502:131–147, 1997.

Stefano Vicini et al. Functional and pharmacological differences between recombinant N-methyl-D-aspartate receptors. *Journal of Neurophysiology*, 79:555–566, 1998.

Huai-Xing Wang, George G. Stradtman, Xiao-Jing Wang, and Wen-Jun Gao. A specialized NMDA receptor function in layer 5 recurrent microcircuitry of the adult rat prefrontal cortex. *Proc Natl Acad Sci USA*, 105:16791–16796, 2008.

Min Wang et al. NMDA receptors subserve persistent neuronal firing during working memory in dorsolateral prefrontal cortex. *Neuron*, 77:736–749, 2013.

Xiao-Jing Wang. Synaptic basis of cortical persistent activity: the importance of NMDA receptors to working memory. *Journal of Neuroscience*, 19:9587–9603, 1999.

Xiao-Jing Wang. Synaptic reverberation underlying mnemonic persistent activity. *Trends in Neurosciences*, 24:455–463, 2001.

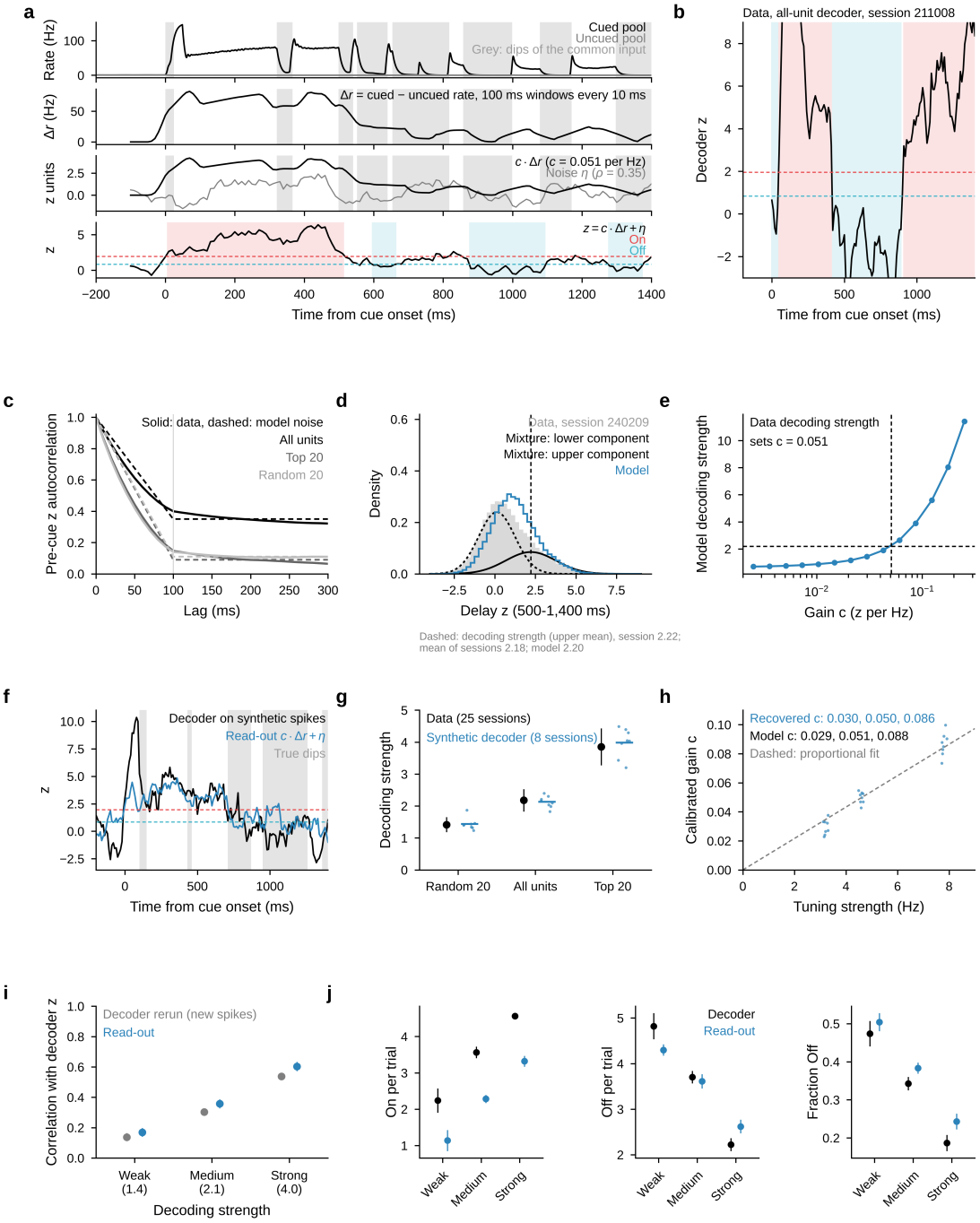
Xiao-Jing Wang. Probabilistic decision making by slow reverberation in cortical circuits. *Neuron*, 36:955–968, 2002.

Yun Wang, Henry Markram, Philip H. Goodman, Thomas K. Berger, Junying Ma, and Patricia S. Goldman-Rakic. Heterogeneity in the pyramidal network of the medial prefrontal cortex. *Nat Neurosci*, 9:534–542, 2006.

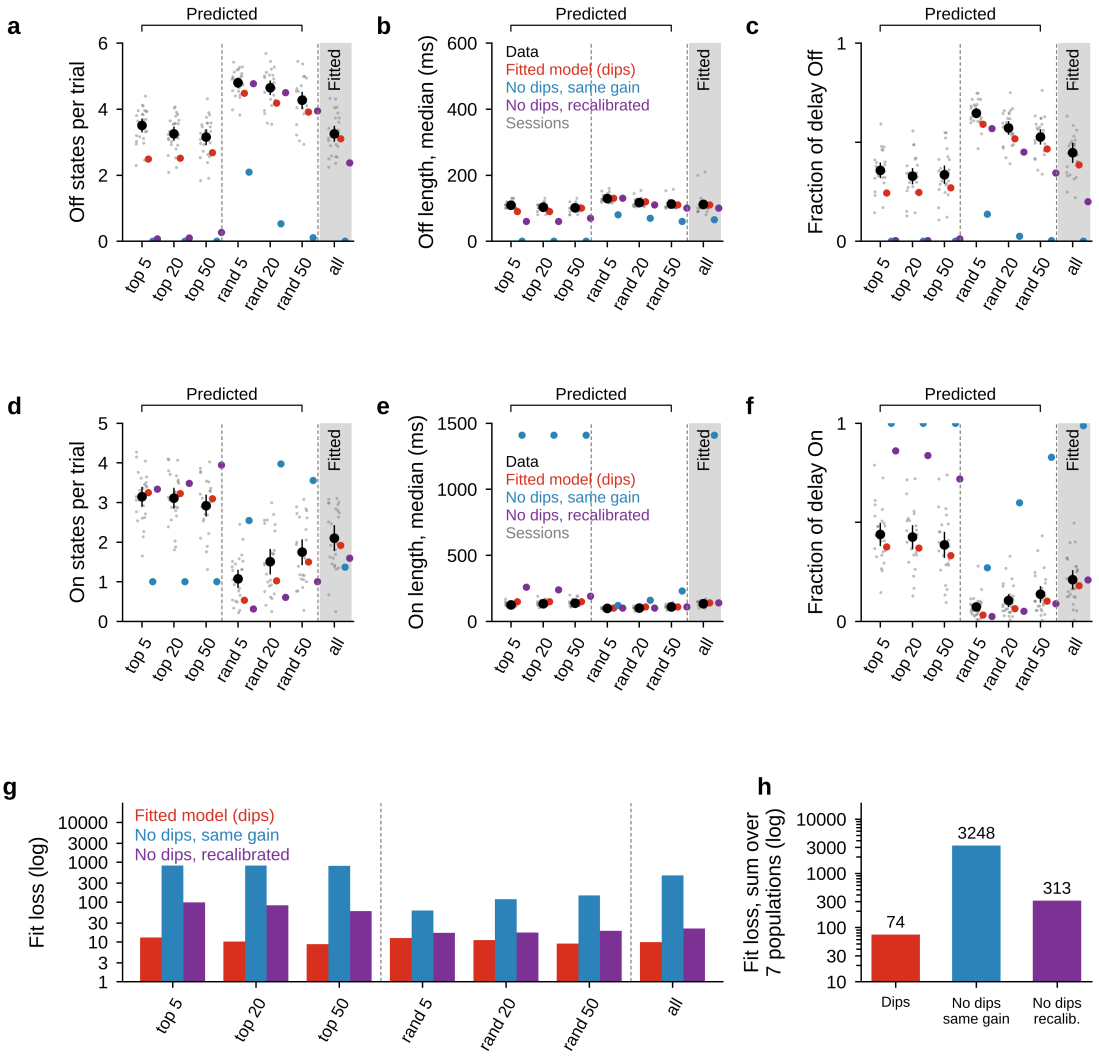
Michael J. Wolff, Janina Jochim, Elkan G. Akyürek, and Mark G. Stokes. Dynamic hidden states underlying working-memory-guided behavior. *Nat Neurosci*, 20:864–871, 2017.

Kong-Fatt Wong and Xiao-Jing Wang. A recurrent network mechanism of time integration in perceptual decisions. *Journal of Neuroscience*, 26:1314–1328, 2006.

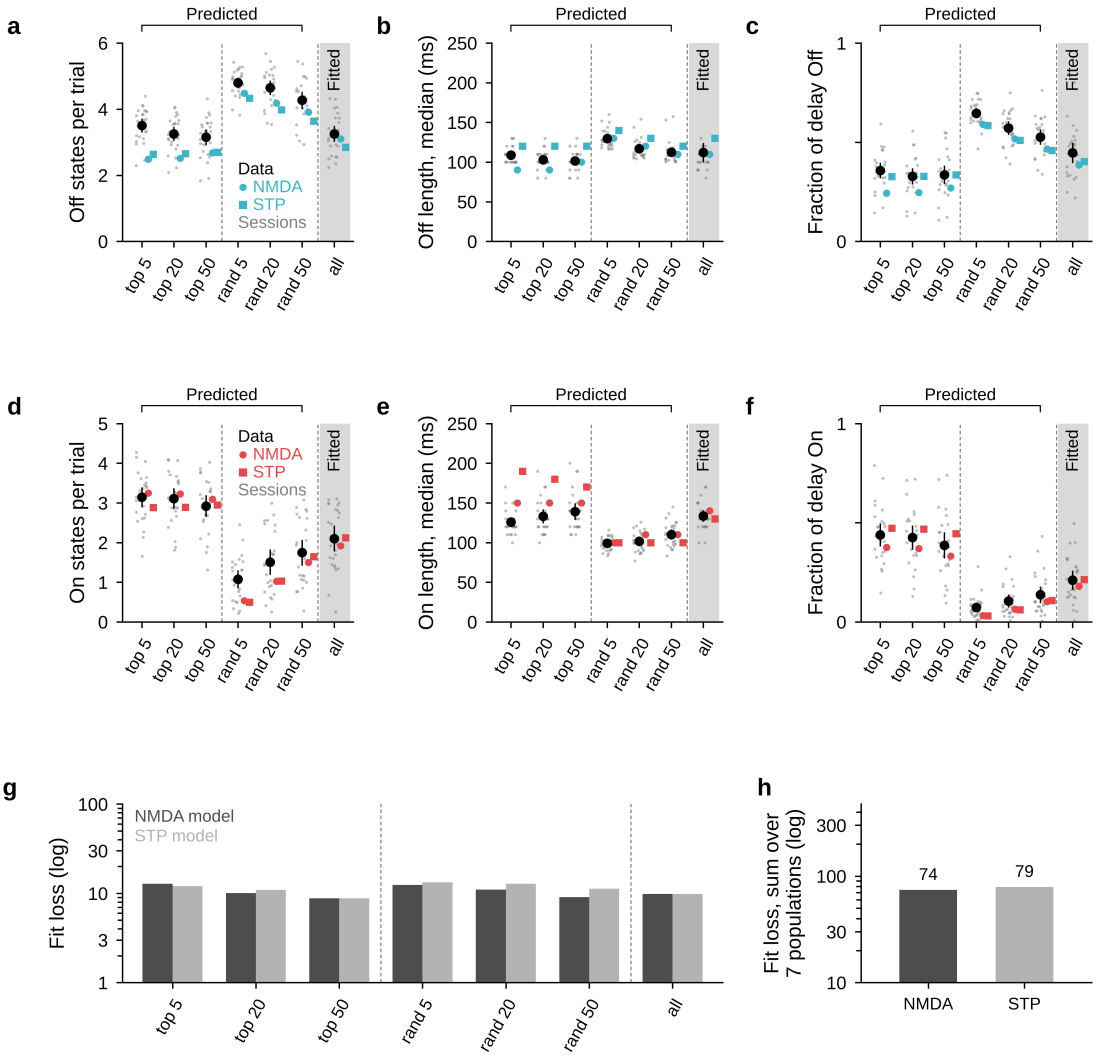
Supplementary Figures



Supplementary Figure 1: Reading the model out like the data. **a**, One model trial through the read-out. From top to bottom, rates of the cued (black) and uncued (grey) pools, with dips of the common input in grey; Δr, their difference averaged in 100 ms windows every 10 ms; $c \cdot \Delta r$ (black) and the read-out noise η (grey); and $z = c \cdot \Delta r + \eta$, with *On* (red) and *Off* (cyan) states labelled with the thresholds of Panichello et al. (2024) (dashed, $z = 1.96$ and 0.84). **b**, One data trial (all-unit decoder, session 211008) with its *On* and *Off* states. **c**, Autocorrelation of the decoder z before the cue, where there is no memory signal, for three decoded populations (solid; black, all units; dark grey, top 20; light grey, random 20), and of the model's noise η with the ρ of each population (dashed). The vertical line marks the 100 ms window, beyond which the correlation reflects the per-trial offset. **d**, Delay z (500–1,400 ms after cue onset) of one example session (grey; session 240209, whose decoding strength is closest to the mean over sessions), with the two components of a Gaussian mixture fitted to it (dotted, lower; solid, upper), and the model's delay z (blue). Dashed, decoding strength, the mean of the upper component (2.22 for this session, 2.18 averaged over sessions, 2.20 for the model). **e**, Decoding strength of the model as a function of the gain c. The data's decoding strength (horizontal line) sets c (vertical line). **f–j**, Validation on synthetic data. The fitted model drove 330 Poisson neurons, at three tuning strengths, and their spikes were decoded with the data pipeline. **f**, Example trial with the true dips (grey), the decoder's z (black) and the read-out $z = c \cdot \Delta r + \eta$ (blue). **g**, Decoding strength of the synthetic decoder (blue, 8 sessions per tuning strength) and of the data's random-20, all-unit and top-20 decoders (black, mean ± 2 SEM over 25 sessions). **h**, Gain c recovered by applying the data calibration to the synthetic decoder, against the tuning strength of the synthetic neurons. Dashed, proportional fit. **i**, Correlation of the read-out's delay z with the decoder's



Supplementary Figure 2: *Off states* are not produced by the read-out noise alone. *On/Off* statistics of the seven decoded populations, as in Fig. 2: **a**, *Off states* per trial; **b**, median *Off* length; **c**, fraction of the delay *Off*; **d**, *On states* per trial; **e**, median *On* length; **f**, fraction of the delay *On*. Black circles, data (mean ± 2 SEM over 25 sessions). Grey dots, single sessions. Red, fitted model (with dips). Blue, the same model with a constant common input (no dips), read out with the identical read-out (same gain, noise and labelling) for each population. Purple, the no-dip model with the read-out gain recalibrated to each population's decoding strength in the data. Without dips, the identical read-out produced no *Off states* for the top-unit and all-unit decoders and few for random neurons. With the gain recalibrated, noise alone reproduced the *Off states* of the random populations but not those of the top-unit decoders (0.1–0.3 vs 3.2–3.5 *Off states* per trial; summed fit loss over the seven populations 313, vs 74 for the fitted model). **g**, Fit loss of each condition for each decoded population (the loss used for fitting, Methods; log scale). **h**, Fit loss summed over the seven decoded populations.



Supplementary Figure 3: The STP model fits the *On/Off* statistics about as well as the NMDA model. As Fig. 2, for the NMDA model (circles) and the STP model, in which short-term facilitation replaces NMDA gating (squares; Methods). Both models were fitted to the decoder using all neurons only (grey band), and the other six decoded populations are predictions obtained by changing only the read-out gain and noise. Black circles, data (mean ± 2 SEM over 25 sessions). Grey dots, single sessions. Cyan and red symbols, model *Off* and *On* statistics. **g**, Fit loss of each model for each decoded population (the loss used for fitting, Methods; log scale). **h**, Fit loss summed over the seven decoded populations.